I keep getting a mismatch error but i don't know why?

No worries! We can figure it out together. 😊

A **mismatch error** in Python often means that two things don't match in the way Python or your assignment expects. It could involve data types, values, variable names, or even expected output.

Let's narrow it down.

First, can you share two things?

1. The exact error message you're getting (copy and paste it if possible).
2. The part of your Python code where the error happens.

I'll help you trace through it rather than just rewrite your code.

One question to start: **Does the error say `TypeError`, `ValueError`, or something like `expected output does not match`?**

We'll take it one step at a time!

Type Mismatch

Your code tried to use a value in a way that does not work for its type (for example, mixing text and numbers).

request.route_execution_failed: Error while processing request for URL 'save_assignment': can only concatenate str (not "int") to str

What To Try Next

- Check that each function call has the right number of arguments.
- Verify that values have the type your code expects (for example, text vs number).
- Print intermediate values to see what type they are before the failing line.

Navigation Options:

- Return to Index Page
- Reset State and Return to Index
- Reload Entire Application
- Submit a Bug Report

Technical Details

Error ID	request.route_execution_failed
Message	Error while processing request for URL 'save_assignment': can only concatenate str (not "int") to str
Route Call	save_assignment(State(assignments=[], course_name='cisc', current_grade=0.0, target_grade=90.0, needed_grade=0.0, message='Course name saved.'), 'exam 1', 70.0)
Status	error
Severity	error
Category	request
Recoverable	yes
Request	Show details

id	4
action	link
url	save_assignment
kwargs	

assignment_name	exam 1
grade	70
--submit-button	"Save Assignment#drafter-component-37418792"

event	{}
dom_id	drafter-component-37418792
button_pressed	drafter-component-37418792
raw_payload	

-

name	assignment_name
value	exam 1

source `form_field`

source_detail

1.

name `grade`

value `70`

source `form_field`

source_detail

1.

name `--submit-button`

value `"Save Assignment#drafter-component-37418792"`

source `form_field`

source_detail

Traceback

Read from the bottom up: the last line names the error, and the nearest “your code” entry above it shows where to look.

Traceback (most recent call last):

File `/lib/python3.14/site-packages/drafter/client_server/client_server.py` , line `523` , in `execute_route`

return route_func(*args, **kwargs), representation ~~~~~~~~~~^^^^^^^^^^^^^^^^^

File `unsaved_thonny_script.py` , line `113` , in `save_assignment` **your code**

return view_assignments(state)

File `unsaved_thonny_script.py` , line `126` , in `view_assignments` **your code**

assignment.name + ":" + int(assignment.grade) + "%"
~~~~~~~~~~~~~~~~~~~~~~^~~~~~~~~~~~~~~~~~~~~~~
~~~~~~~~~~~~~~~~~~~~~~

TypeError: can only concatenate str (not "int") to str

Route:save_assignment❌ 1📌 Persisted (0)🖨️ Console

🖨️ **Console**ClearHide

Pillow already loaded from default channel Loading Jinja2, MarkupSafe, contourpy, cycler, fonttools, kiwisolver, matplotlib, numpy, packaging, pyparsing, python-dateutil, pytz, six Loaded Jinja2, MarkupSafe, contourpy, cycler, fonttools, kiwisolver, matplotlib, numpy, packaging, pyparsing, python-dateutil, pytz, six TEST PASSED - [line 169] assert_equal(average_grades(test_assignments), 80.0) TEST PASSED - [line 170] assert_equal(average_grades([]), 0.0)

>>>

Debug Panel

View

Current1HistoryOverviewTestsEnvironment

Current Page

Route: `save_assignment`

Page Content

Errors & Warnings

❌Your code tried to use a value in a way that does not work for its type (for example, mixing text and numbers).Details

🔍 Compare with previous state

Current State

State

assignments:

[]

Empty list

course_name:

'cisc'

str

current_grade:

0.0

float

target_grade:

90.0

float

needed_grade:

0.0

float

message:

'Course name saved.'

str

Great, this traceback gives us a specific place to investigate. The important part is:

```
None
assignment.name + ":" + int(assignment.grade) + "%"
```

The error says:

`can only concatenate str (not "int") to str`

What Python is telling you

Python is evaluating the `+` operators from left to right here.

- `assignment.name` is likely a string.
- `":"` is a string.
- `int(assignment.grade)` produces an integer.

- `"%"` is a string.

The issue occurs when Python tries to combine a string with an integer using `+`. The `+` operator can concatenate strings, but it cannot directly concatenate a string and an integer.

Your textbook explains that **types determine which operations are allowed**. For example, strings can be joined with other strings, but you cannot subtract strings or combine a string and a number through string concatenation.

Let's reason it through

Before changing anything, consider these questions:

1. What type does `int(assignment.grade)` produce?
2. What type does Python expect if you're using `+` to join that value to `assignment.name`?
3. Do you want the grade to be used as a number for a calculation, or displayed as text on the page?

One more question: What is the purpose of `view_assignments` in your program—are you trying to display each assignment's name followed by its grade and a percent sign?

Your answers will help us decide what needs to change without accidentally affecting the grade calculations elsewhere.

```
from dataclasses import dataclass
from drafter import *
from bakery import assert_equal


@dataclass
class Assignment:
    """Store the name and grade for one assignment."""
    name: str
    grade: float


@dataclass
class State:
    """Store information that the website needs between pages."""
    assignments: list[Assignment]
    course_name: str
```

```
    current_grade: float
    target_grade: float
    needed_grade: float
    message: str


def average_grades(assignments: list[Assignment]) -> float:
    """Calculate the average of all assignment grades."""
    if len(assignments) == 0:
        return 0.0

    total = 0.0
    for assignment in assignments:
        total = total + assignment.grade

    return total / len(assignments)


def required_next_grade(assignments: list[Assignment],
                        target: float) -> float:
    """Calculate the grade needed on the next assignment."""
    total = 0.0

    for assignment in assignments:
        total = total + assignment.grade

    number_of_assignments = len(assignments) + 1
    needed = target * number_of_assignments - total

    return needed


def valid_grade(grade: float) -> bool:
    """Check whether a grade is between 0 and 100."""
    return grade >= 0 and grade <= 100


@route
def index(state: State) -> Page:
    """Display the home page and main navigation."""
    return Page(state, [
        "COLLEGE GRADE CALCULATOR",
        "Welcome! Use this website to keep track of your grades.",
        "Course name: " + state.course_name,
```

```
        TextBox("course_name", state.course_name),
        Button("Save Course Name", "save_course"),
        " ",
        Button("Add Assignment", "add_assignment"),
        Button("View Assignments", "view_assignments"),
        Button("Calculate Current Grade", "calculate_grade"),
        state.message
    ])


@route
def save_course(state: State, course_name: str) -> Page:
    """Save the course name entered by the student."""
    if course_name.strip() == "":
        state.message = "Please enter a course name."
    else:
        state.course_name = course_name
        state.message = "Course name saved."

    return index(state)


@route
def add_assignment(state: State) -> Page:
    """Display the form for entering a new assignment."""
    return Page(state, [
        "ADD AN ASSIGNMENT",
        "Enter the assignment name:",
        TextBox("assignment_name"),
        "Enter the assignment grade (0-100):",
        TextBox("grade", 0.0, "number"),
        Button("Save Assignment", "save_assignment"),
        Button("Home", "index")
    ])


@route
def save_assignment(state: State, assignment_name: str,
                    grade: float) -> Page:
    """Check and save a new assignment."""
    if assignment_name.strip() == "":
        state.message = "Please enter an assignment name."
    elif not valid_grade(grade):
        state.message = "Grade must be between 0 and 100."
```

```
    else:
        new_assignment = Assignment(assignment_name, grade)
        state.assignments.append(new_assignment)
        state.current_grade = average_grades(state.assignments)
        state.message = "Assignment successfully added."

    return view_assignments(state)


@route
def view_assignments(state: State) -> Page:
    """Display all assignments currently saved."""
    content = ["YOUR ASSIGNMENTS"]

    if len(state.assignments) == 0:
        content.append("You have not entered any assignments yet.")
    else:
        for assignment in state.assignments:
            content.append(
                assignment.name + ":" + int(assignment.grade) + "%"
            )

    content.append(" ")
    content.append("Current average: " +
                   int(round(average_grades(state.assignments), 2)) + "%")
    content.append(state.message)
    content.append(Button("Add Another Assignment", "add_assignment"))
    content.append(Button("Calculate Current Grade", "calculate_grade"))
    content.append(Button("Home", "index"))

    return Page(state, content)


@route
def calculate_grade(state: State) -> Page:
    """Calculate and display the student's current grade."""
    if len(state.assignments) == 0:
        state.message = "Add assignments before calculating your grade."
        state.current_grade = 0.0
    else:
        state.current_grade = average_grades(state.assignments)
        state.message = "Your current grade has been calculated."

    return Page(state, [
```

```
        "YOUR CURRENT GRADE",
        "Course: " + state.course_name,
        "Current average: " + str(round(state.current_grade, 2)) + "%",
        state.message,
        Button("View Assignments", "view_assignments"),
        Button("Home", "index")
    ])



# Tests for the calculation and validation functions
test_assignments = [
    Assignment("Homework 1", 80.0),
    Assignment("Quiz 1", 90.0),
    Assignment("Exam 1", 70.0)
]

assert_equal(average_grades(test_assignments), 80.0)
assert_equal(average_grades([]), 0.0)


start_server(State([], "", 0.0, 90.0, 0.0, ""))
```

Thanks for sharing your code! I can see exactly where the error is coming from. Let's look at the types involved rather than rewriting the function for you.

1. The first mismatch

Inside `view_assignments`, you have:

```
None
assignment.name + ":" + int(assignment.grade) + "%"
```

Think about the types of each part:

Expression	Type

`assignment.name`	`str`
`":"`	`str`
`int(assignment.grade)`	`int`
`"%"`	`str`

Python's `+` operator concatenates strings, but it cannot concatenate a string and an integer. That's why you see the `TypeError`.

2. There's another similar mismatch

A little farther down, you have:

```
None
"Current average: " + int(round(average_grades(state.assignments), 2)) + "%"
```

Let's trace the types:

- `"Current average: "` is a string.
- `average_grades(...)` returns a float.
- `round(..., 2)` returns a number.
- `int(...)` converts that number to an integer.

So this expression has the same kind of mismatch.

3. Your turn: reason about the conversion

You already use `str()` successfully in `calculate_grade`:

```
None
"Current average: " + str(round(state.current_grade, 2)) + "%"
```

Compare that expression with the two expressions in `view_assignments`.

A couple of questions:

1. What does `str()` do to a numeric value?
2. Why might `str()` be more appropriate than `int()` when you want to display a grade such as `87.5%`?
3. Why would converting a grade to an integer potentially lose information?

The textbook's section on types explains that a value's type determines which operations Python allows. Here, the important distinction is between a number used for calculations and text used for display.

Try making your own change to the two expressions in `view_assignments`. Then test the page by saving an assignment with a grade of `70.5`. What do you observe?

Annotations

1 error, 3 warnings, and 2 notices

build	**Process completed with exit code 1.**
build	Node.js 20 is deprecated. The following actions target Node.js 20 but are being forced to run on Node.js 24: astral-sh/setup-uv@v6. For more information see: https://github.blog/changelog/2025-09-19-deprecation-of-node-20-on-github-actions-runners/
build	No file matched to [/home/runner/work/college-course-grader/college-course-grader/**/*requirements*.txt,/home/runner/work/college-course-grader/college-course-grader/**/*requirements*.in,/home/runner/work/college-course-grader/college-course-grader/**/*constraints*.txt,/home/runner/work/college-course-grader/college-course-grader/**/*constraints*.in,/home/runner/work/college-course-grader/college-course-grader/**/pyproject.toml,/home/runner/work/college-course-grader/college-course-grader/**/uv.lock,/home/runner/work/college-course-grader/college-course-grader/**/*.py.lock]. The cache will never get invalidated. Make sure you have checked out the target repository and configured the cache-dependency-glob input correctly.
deploy	Node.js 20 is deprecated. The following actions target Node.js 20 but are being forced to run on Node.js 24: actions/deploy-pages@v4. For more information see: https://github.blog/changelog/2025-09-19-deprecation-of-node-20-on-github-actions-runners/

build	"The ubuntu-latest label will migrate to Ubuntu 26 beginning October 19, 2026. For more information, see https://github.com/actions/runner-images/issues/14748"
deploy	"The ubuntu-latest label will migrate to Ubuntu 26 beginning October 19, 2026. For more information, see https://github.com/actions/runner-images/issues/14748"

Artifacts

Good news: these annotations show several warnings, but **they don't all indicate problems with your Python code**. Let's separate them.

1. The important message: build failed

Build failed

The key line is:

```
Process completed with exit code 1
```

This tells us that the build process failed, but it does not tell us why. We need the actual error earlier in the build log.

2. The other annotations

Annotation	What it means
Node.js 20 is deprecated	A GitHub Actions maintenance warning.
No file matched requirements	The dependency cache couldn't find a matching requirements or project configuration file.
Ubuntu 26 migration	A notice about a future runner change.

These messages are not, by themselves, proof of a Python error.

3. Let's investigate the actual failure

Your earlier Python error happened in `view_assignments`, but these GitHub Actions annotations don't show whether that issue is still present.

Open the failed **build** job and look at the log around the first red `Error` message, especially steps after the dependency setup.

Can you paste the first actual error message from the build log?

One useful distinction to keep in mind: a warning is not necessarily a failure. We want to find the message that explains why the build exited with code 1, rather than changing things based on unrelated warnings.

build **Node.js 20 is deprecated. The following actions target Node.js 20 but are being forced to run on Node.js 24: astral-sh/setup-uv@v6. For more information see: https://github.blog/changelog/2025-09-19-deprecation-of-node-20-on-github-actions-runners/**

This message is a **warning, not the cause of your build failure**.

It means that the GitHub Actions workflow is using `astral-sh/setup-uv@v6`, which targets Node.js 20. GitHub is currently forcing it to run on Node.js 24 because Node.js 20 is deprecated.

What you should know

- It does not identify an error in your Python code.
- It does not mean your grade calculator is broken.
- It is a GitHub Actions maintenance warning.
- You don't need to change your Python functions to address this warning.

Let's find the real problem

The build previously reported `exit code 1`, which means something caused the build to fail.

Can you open the build job and share the output from the step immediately before the build fails?

Look for a red ❌ or a message containing `Error`, `Failed`, or `Traceback`.

I don't want you changing working code to fix an unrelated warning.

Run set -euo pipefail
 Drafter v2 deployment build
 Repository: gavinbel-lgtm/college-course-grader

Run: https://github.com/gavinbel-lgtm/college-course-grader/actions/runs/43481946812
Entry file: main.py
Command: uvx --from drafter\>=2\,\<3 drafter main.py --compile --output-directory _site --production --additional-paths Copy\ of\ CS1\ Final\ Project\ 0_\ Pick\ a\ Problem.pdf --additional-paths Copy\ of\ CS1\ Final\ Project\ 1_\ Design\ \(1\).pdf --additional-paths README.md --additional-paths project\ LLM\ \(1\).pdf --external-pages dashboard/\ Dashboard --external-pages https://github.com/gavinbel-lgtm/college-course-grader\ Repository
Installed 39 packages in 22ms
Traceback (most recent call last):
File "/home/runner/work/_temp/setup-uv-cache/archive-v0/YZbBtr1EGcHaTp_5/bin/drafter", line 12, in <module>
sys.exit(main())
^^^^^^
File "/home/runner/work/_temp/setup-uv-cache/archive-v0/YZbBtr1EGcHaTp_5/lib/python3.12/site-packages/drafter/cli.py", line 34, in main
run_path(system.bootstrap.path, run_name="__main__")
File "<frozen runpy>", line 286, in run_path
File "<frozen runpy>", line 259, in _get_code_from_file
File "main.py", line 87
Calculate the grade needed on the next assignment.
^^^
SyntaxError: invalid syntax
Error: Process completed with exit code 1.

Excellent! We have found the actual cause of the build failure. This time, it is a **Python syntax error**, not a type mismatch.

The most important lines are:

```
None
File "main.py", line 87
    Calculate the grade needed on the next assignment.
              ^^^
SyntaxError: invalid syntax
```

1. What does this mean?

Python is trying to interpret the words:

```
Calculate the grade needed on the next assignment.
```

as Python code, but they aren't valid Python statements.

Look at the function you shared earlier:

```
None
def required_next_grade(assignments: list[Assignment],
                        target: float) -> float:
    """Calculate the grade needed on the next assignment."""
```

Notice the difference between a comment or docstring and ordinary text.

- A docstring is enclosed in triple quotation marks (`"""`).
- A comment begins with `#`.
- Ordinary English text by itself is not a valid Python statement.

2. Let's investigate line 87

I want you to check your actual `main.py` file.

Question: Does line 87 contain the sentence `Calculate the grade needed on the next assignment.` without quotation marks or a `#`?

It may be that your docstring's opening triple quotation marks were accidentally removed or misplaced.

3. One important distinction

You now have two separate issues to keep track of:

Error	Meaning
Earlier `TypeError`	Mixing strings and integers with `+`.
Current `SyntaxError`	Python cannot understand the syntax around line 87.

Let's resolve the syntax error first.

Can you paste lines 80–95 from your current `main.py` file? That will help us identify exactly what Python is interpreting incorrectly.

I found the exact cause! Your uploaded `main.py` confirms the syntax error reported by GitHub Actions.

1. The problem is in `required_next_grade`

Look at this section of your file:

```
None
def required_next_grade(assignments: list[Assignment],
                        target: float) -> float:
    """
    Calculate the grade needed on the next assignment.

    Args:
        assignments: A list of completed Assignment objects.
        target: The desired overall grade.

    Returns:
        The grade needed on the next equally weighted assignment.
    """

    Calculate the grade needed on the next assignment.
    total = 0.0
```

Notice something?

You already have a complete docstring describing the function. But immediately after the closing triple quotation marks, there's another sentence:

`Calculate the grade needed on the next assignment.`

Python is interpreting that sentence as executable code.

2. Your debugging question

Compare these two lines:

- The sentence inside the triple quotation marks.
- The identical sentence outside the triple quotation marks.

Why do you think Python accepts the first one but rejects the second one?

3. Check your understanding

Your task is to identify what should happen to that extra sentence so that Python no longer tries to execute it as code.

After making your change, run the program again.

4. One encouraging observation

I also noticed that your `view_assignments` function now uses `str()` when displaying grades:

```
None
assignment.name + ": " + str(assignment.grade) + "%"
```

That addresses the string/integer concatenation issue we discussed earlier.

Your grade calculation functions also have tests, including tests for empty assignment lists and required grades.

Let's fix the syntax issue first, then see what the next build tells us.

Run git log --oneline --decorate --graph --all -n 50 > _site/dashboard/commits.txt
/home/runner/work/_temp/f0cc79b6-9cbb-4048-a5d4-76fc31e64f21.sh: line 1: _site/dashboard/commits.txt: No such file or directory
Error: Process completed with exit code 1.

Good, this is a different error from the Python errors we were investigating. This time, the failure is happening in your **GitHub Actions workflow**, not inside your grade calculator.

1. What the error means

The important message is:

```
None
_site/dashboard/commits.txt: No such file or directory
```

Your workflow is attempting to run:

```
None
git log --oneline --decorate --graph --all -n 50 >
_site/dashboard/commits.txt
```

The `>` operator redirects the Git log output into a file.

However, the operating system cannot create that file because the directory `_site/dashboard/` does not exist at that moment.

2. Understand the distinction

Think of a directory as a folder:

- `_site` is one folder.
- `dashboard` is supposed to be a folder inside `_site`.
- `commits.txt` is the file you want to create inside `dashboard`.

Creating a file does not automatically create its parent directories.

3. Let's investigate your workflow

Before changing anything, I want you to check:

1. Does your repository contain a `dashboard` folder?
2. Does the Drafter build create `_site/dashboard/`, or is that directory supposed to be created by another workflow step?
3. Does the GitHub Actions step that runs `git log` happen before or after the Drafter build?

Here's a useful hint: In a shell, `mkdir -p` is a command that creates a directory and its parent directories if necessary.

Can you share the section of your GitHub Actions YAML file containing the `git log` command?

We'll determine whether this is an ordering issue or a missing-directory issue before changing your deployment workflow.

Run set -o pipefail
main.py:4: error: Name "assert_equal" already defined (possibly by an import) [no-redef]
main.py:238: error: Argument 1 to "append" of "list" has incompatible type "Button"; expected "str" [arg-type]
main.py:239: error: Argument 1 to "append" of "list" has incompatible type "Button"; expected "str" [arg-type]
main.py:240: error: Argument 1 to "append" of "list" has incompatible type "Button"; expected "str" [arg-type]
Found 4 errors in 1 file (checked 1 source file)
Error: Process completed with exit code 1.

Great, this log gives us four specific errors from your Python type checker (likely mypy). These are different from the earlier syntax and runtime errors.

Let's examine them one at a time.

1. Error on line 4: `no-redef`

```
None
main.py:4: error: Name "assert_equal" already defined
(possibly by an import)
```

You have this import in your program:

```
None
from bakery import assert_equal
```

The type checker is telling you that `assert_equal` appears to be defined more than once.

Think about this: Do you have another function or variable named `assert_equal` somewhere in your file, or is it being imported from another module?

2. Errors on lines 238–240: `arg-type`

```
None
Argument 1 to "append" of "list" has incompatible type "Button";
expected "str"
```

These three errors are related.

Look at how you originally created your content list:

```
None
content = ["YOUR ASSIGNMENTS"]
```

Python infers that this is a list of strings (`list[str]`).

Later, your code uses:

```
None
content.append(Button(...))
```

But a `Button` is not a string.

Your textbook's discussion of types is relevant here: types determine which values and operations are valid.

3. Let's reason through the list

Consider these questions:

1. What type of value does `content` initially contain?
2. What type of object does `Button(...)` produce?
3. Why does the type checker complain when you try to append a Button to a list it expects to contain strings?

4. Your next debugging step

I don't want to rewrite your code for you. Let's establish what the intended design is first.

Question: In your Drafter application, does the `Page` content need to contain both ordinary text and interface components such as `Button` objects?

That distinction will help us understand what type the content list should represent.

Once we resolve these errors, we can run the type checker again and see whether anything else needs attention.